

```

import subprocess
import sys

try:
    import cv2
except ImportError:
    print("[INFO] OpenCV not found. Installing opencv-python
automatically...")
    subprocess.check_call(
        [sys.executable, "-m", "pip", "install", "opencv-python"]
    )
    import cv2

def run_facial_recognition():
    face_cascade = cv2.CascadeClassifier(
        cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
    )

    video_capture = cv2.VideoCapture(0)

    if not video_capture.isOpened():
        print("[ERROR] Could not open webcam.")
        return

    print("Facial Recognition active. Press 'q' to quit the
application.")

    while True:
        ret, frame = video_capture.read()
        if not ret:
            print("[ERROR] Failed to grab frame.")
            break

        gray_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

        faces = face_cascade.detectMultiScale(
            gray_frame, scaleFactor=1.1, minNeighbors=5, minSize=(30,
30)
        )

        for x, y, w, h in faces:
            cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0),
2)

            cv2.putText(
                frame,
                "Face Detected",
                (x, y - 10),

```

```
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (0, 255, 0),
        2,
    )

    cv2.imshow("Facial Recognition System", frame)

    if cv2.waitKey(1) & 0xFF == ord("q"):
        break

    video_capture.release()
    cv2.destroyAllWindows()
    print("Application closed successfully.")

if __name__ == "__main__":
    run_facial_recognition()
```